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FAILURE DETECTION AND RECOVERY IN ROBOTIC ARM GRASP

BEng Mechatronics and Robotics

Major Project (PDE3823)

Technical Report

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Chapter 1: Introduction

1.1 Background and Motivation

Robotic manipulation is central to modern industrial automation, with pick-and-place operations deployed across manufacturing, warehousing, pharmaceutical handling, and logistics. Despite decades of progress in robot hardware and control, grasp failure remains a persistent problem. Reported grasp failure rates in industrial bin-picking tasks typically range from 5% to 25%, depending on object geometry, surface properties, and the accuracy of scene reconstruction [1]. Even at the lower end of this range, a system executing thousands of picks accumulates a significant number of failures, each of which carries a cost in downtime, throughput loss, or damaged goods.

The research community has largely approached this in two ways: making grasps more likely to succeed in the first place, or detecting when one has failed. Both are useful, but neither solves the operational problem on its own. Zhu et al. [2] tackled this directly, building a system that detects failures at every phase of a pick-and-place task and recovers from them automatically by combining camera-based tracking with force feedback from a pneumatic soft gripper. Their results show a substantial improvement in task completion without human intervention. The limitation is that their approach depends on a pneumatic soft gripper, which measures internal air pressure to sense contact and slip. The majority of deployed industrial grippers are rigid parallel-jaw devices that offer only binary force-resistant signals and jaw width feedback, providing substantially less information about grasp state. Extending closed-loop failure handling to this class of gripper, without the richer proprioceptive signal, is the problem this project addresses.

1.2 Problem Statement

Given a rigid parallel-jaw gripper with limited proprioceptive feedback, how can the system determine with high confidence whether a grasp attempt succeeded, classify the type of failure when it did not, and execute an appropriate corrective action to complete the task without human assistance?

Three failure modes are investigated. The first is a position shift failure, where the target object is not at the expected pose when the gripper closes. This includes cases where the object moved before approach, where calibration error placed the gripper off-target, and where the gripper made lateral contact with the object without enclosing it, producing a false positive on both the DI8 signal and jaw width reading. The second is an orientation failure, specific to cylindrical objects resting horizontally rather than upright. The RG2 maximum jaw stroke of 110 mm cannot reliably encircle a horizontal cylinder without a

corrected approach orientation. The third is a slip failure, where the object is initially secured but is lost during the transport phase due to insufficient grip force.

These failure modes reflect conditions consistently reported in semi-structured industrial deployments where object placement and orientation are not guaranteed. A system that handles all three autonomously represents a meaningful improvement in the operational robustness of rigid-gripper manipulation.

1.3 Aims and Objectives

The overall aim is to design, implement, and experimentally evaluate a grasp failure detection and recovery system on a UR10 CB-Series robot arm with an OnRobot RG2 gripper and OAK-D Lite stereo depth camera mounted in an eye-in-hand configuration.

Objective 1 — Multi-Modal Sensing Architecture

Implement a sensor fusion pipeline combining the RG2 DI8 signal and A12 jaw-width reading with a camera-based post-grasp verification layer. The gripper sensors serve as a first-pass trigger; the camera provides definitive confirmation or refutation of grasp success, resolving false positives the gripper alone cannot distinguish.

Validation: DI8 false positive rate documented from 30 induced-failure baseline trials before the camera layer is added.

Target: the fused pipeline reduces the DI8 false positive rate by at least 50%.

Objective 2 — Binary Post-Grasp Classifier

Develop and evaluate a Binary Gripper State Classification method using the OAK-D Lite mounted at the wrist at approximately 35 degrees downward, capturing both the floor workspace pre-grasp and the gripper jaws post-grasp in a single frame. A YOLOv8n-cls binary classifier determines HOLDING or EMPTY from a gripper region-of-interest crop after lifting 200 mm.

Validation: classifier accuracy, precision, recall, and F1 on a held-out test set of 60 labelled images (30 HOLDING, 30 EMPTY).

Target: overall accuracy at least 85% $((TP + TN) / 60)$, true positive rate at least 90%.

Objective 3 — Type-Specific Recovery Routines

Implement a distinct recovery routine for each failure mode. For position shift and nudge failures: re-detect the object at its updated pixel location and execute a corrected grasp from the new position. For orientation failure: use the OBB angle extracted from YOLOv8 OBB detection to compute a corrected gripper rotation ($R_z = R_{z_default} +$

OBB_angle + 90°), aligning the jaws perpendicular to the cylinder axis before re-attempting. For slip: re-detect the dropped object and re-grasp with increased grip force.

Validation: recovery success rate per failure type across the 90-trial protocol.

Targets: at least 70% for nudge recovery, at least 70% for orientation recovery, at least 60% for slip recovery.

Objective 4 — Structured Experimental Evaluation

Evaluate the complete system across 90 trials, 30 per failure type, run in two configurations: detection-only (recovery disabled, isolating classification accuracy) and full system (detection and recovery both active, end-to-end performance). Ground truth is recorded manually by the experimenter for every trial, independently of the system output, providing the reference for confusion matrix analysis.

Primary metrics and targets

Metric	Formula	Target	Interpretation
True Positive Rate (TPR)	$TP / (TP + FN)$	$\geq 85\%$	Successful grasps correctly confirmed
True Negative Rate (TNR)	$TN / (TN + FP)$	$\geq 85\%$	Failed grasps correctly identified
False Positive Rate (FPR)	$FP / (FP + TN)$	$< 20\%$	System says success when grasp failed
Overall detection accuracy	$(TP + TN) / 90$	$\geq 85\%$	Combined correct verdicts across all trials
Recovery success rate	Recoveries succeeded / Recoveries attempted	$\geq 70\%$ overall	Proportion of detected failures resolved by recovery
End-to-end task completion rate	Successful placements / 90	$\geq 70\%$ with recovery enabled	Overall system performance

Table 1: Primary experimental metrics across the 90-trial protocol

1.4 Project Scope and Constraints

This project operates within the constraints of the available laboratory infrastructure. The robot platform is a UR10 CB-Series six degree-of-freedom arm with 1300 mm reach and 10 kg payload, controlled via the `ur_rtde` library. The end-effector is an OnRobot RG2 electrically actuated parallel-jaw gripper providing DI8 and A12 jaw-width feedback only, with no native force measurement. The vision system is an OAK-D Lite stereo depth camera mounted in an eye-in-hand configuration at approximately 35 degrees downward, positioned to view both the floor pick zone during approach and the gripper jaws during post-grasp verification.

The scope is limited to three failure types and two object classes: small styrofoam cubes and cylinders of approximately 25 to 85 mm height. Slip detection via force-sensitive resistors is treated as a stretch objective given that the test objects (approximately 30 g) may produce contact forces within sensor noise margins, and the viability of this sensing modality is explicitly evaluated and reported rather than assumed.

Chapter 2: Technical Background and Literature Review

2.1 Introduction

This chapter establishes the technical foundation for the design decisions made in this project and positions the work within the existing research literature. It covers the technical background relevant to this project across four areas: parallel-jaw gripper mechanics and sensing constraints, grasp failure mode taxonomy, published detection and verification approaches, and recovery strategies. It concludes with a critical synthesis identifying the contribution of this project relative to prior work.

2.2 Robotic Grasping: Fundamentals and Challenges

2.2.1 Grasp Planning and Execution

A robotic grasp requires positioning the end-effector such that contact points on the object produce sufficient normal and frictional force to hold it through the lift and transport[3]. For a parallel-jaw gripper this means the approach vector needs to align with the object geometry so both jaws make stable contact at the same time. In controlled environments with known object models and well-calibrated sensors, planning tools like GraspIt! [4] and Dex-Net [5] handle this reliably. In practice, things are less clean: objects are not always where they were detected, camera-to-robot calibration drifts over time, and surface conditions affect how much friction the contact actually provides.

The UR10 used in this project has a positional repeatability of 0.05 mm [6], which is more than sufficient for the objects being handled. The RG2 gripper gives jaw width feedback to millimetre precision and a binary DI8 signal that indicates whether something is detected between the jaws. These define the proprioceptive boundary within which the detection and recovery strategy has to operate.

2.2.2 The False Positive Problem in Binary Gripper Feedback

A well-documented limitation of binary gripper feedback is the false positive condition in which the gripper signals object detection, although the object has not been securely grasped. This occurs when the gripper jaws make lateral contact with an object during approach and nudging it rather than enclosing it [8]. The nudge false positive is especially problematic in bin-picking and floor-level picking scenarios, where objects may shift slightly during gripper descent.

Corrigan et al. [6] demonstrated experimentally that binary gripper signals alone are insufficient for confident grasp verification in more than 20% of failure events in a standard bin-picking task, because the signal cannot distinguish between a secure grasp and a surface contact without enclosure. Zhu et al. [2] ran into a similar problem in their multi-modal system, where soft-gripper pressure signals could detect contact and contact loss but could not reliably tell apart a stable grasp from an unstable pinch that would fail during transport. Their solution was to bring visual tracking in during the lift phase to catch what the pressure signal missed. These findings directly motivate the multi-modal approach taken in this project, where camera-based post-grasp verification is used to resolve the ambiguity that binary gripper feedback cannot.

2.2.3 Camera Geometry for Floor-Level Picking

Mounting the camera at the robot wrist in an eye-in-hand configuration means the camera moves with the arm and keeps a consistent view of the gripper jaws across all poses. By angling the camera approximately 35 degrees downward, both the floor pick zone and the gripper jaws fall within the field of view simultaneously. This lets the same camera handle two roles: object detection and pose estimation during the approach phase, and binary gripper state classification during the post-grasp verification phase, without needing the robot to move to a separate check pose between steps.

The known drawback of eye-in-hand setups is gripper self-occlusion during the final contact phase, where the jaws partially block the object as they close [7]. This is a well-documented trade-off, and one review notes that no camera configuration entirely avoids occlusion [8]. In this project, the classifier is trained on images collected at the post-grasp lift height above the pick point, where an object held in the jaws is visible between or below the jaw faces. The classifier training set explicitly includes partial-occlusion examples to improve robustness to this condition.

A further consequence of the eye-in-hand configuration is that passive stereo depth from the OAK-D Lite becomes unreliable at very short distances. The object held in the jaws is approximately 200 to 400 mm from the wrist camera, which is below the sensor's minimum

reliable depth range. Post-grasp verification therefore relies exclusively on the RGB image and the binary classifier, not on depth data. This is a deliberate choice rather than a workaround since depth is not needed to answer a binary question.

Additionally, Oak D Lite uses matching features across a pair of cameras to estimate depth. This approach requires texture in the scene. Plain, uniform surfaces produce unreliable depth estimates due to the absence of matchable features [22]. The floor-level pick zone in this project uses a textured mat to address this, ensuring depth reliability at the 600–900 mm operating range.

2.3 Taxonomy of Grasp Failure Modes

2.3.1 Position Failures

A position failure occurs when the target object is not at the expected pose when the gripper arrives at the planned grasp point. This happens when the object shifted before the approach due to vibration or disturbance, when kinematic error placed the TCP off target, or when calibration drift between the camera and robot frames meant the planned pick point no longer matched the actual object position. A nudge failure is a specific and practically important subset where the jaws make lateral surface contact with the object and displace it, while DI8 and jaw width both report what appears to be a successful grasp. There is no way to distinguish a nudge from a real grasp using gripper signals alone, because both conditions produce identical readings [2].

Zeng et al. [9] classify these as pre-contact failures since the gripper never achieves the intended enclosing contact. Detection relies on either tracking the object during the approach using CSRT [10] to identify unexpected displacement before the gripper commits, or on post-grasp camera verification to confirm the outcome after the lift.

2.3.2 Orientation Failures

Orientation failures occur when the object is not in the expected pose relative to the gripper approach axis. A cylinder standing upright presents a circular cross-section to an overhead gripper, and closure within the RG2 jaw span is straightforward. For a cylinder lying horizontally, the 110 mm RG2 maximum stroke is not enough to encircle it along its length.

Orientation is estimated from the downward-angled camera using bounding box aspect ratio: an upright cylinder produces a near-square bounding box, while a horizontal one produces an elongated box with an aspect ratio in the range 2.5 to 4.0 depending on cylinder dimensions. YOLOv8 OBB detection extends this by outputting the in-plane rotation angle directly, so the gripper rotation correction can be computed without a separate estimation step. Deng et al. [11] describe point cloud surface normal analysis as an alternative, but this requires reliable depth at close range, which is not available in the eye-in-hand configuration used here.

2.3.3 Slip and In-Transit Failures

Slip occurs after an apparently successful grasp when the object is lost during transport. The contributing factors are insufficient grip force relative to object mass, reduced friction at the contact surface, and dynamic loading during the acceleration phases of the lift trajectory. Detection requires either continuous contact force monitoring or visual observation of the object during transport.

Zhu et al. [2] address this by monitoring internal pressure in their pneumatic gripper bellows during the lift: a pressure drop while the gripper is nominally closed indicates object loss. The RG2 has no equivalent signal. Force-sensitive resistors on the jaw faces can approximate contact force monitoring, but at the test object mass of approximately 30 g, the expected readings sit close to sensor noise margins, as discussed in Section 1.4. The camera-based alternative tracks the object's vertical centroid position during the lift and flags a downward drift as a slip event, though this only works if the object remains within the camera field of view throughout the trajectory.

2.4 Post-Grasp Verification Approaches

2.4.1 Proprioceptive Verification and Its Limits

Jaw width monitoring is the most commonly deployed verification approach in industrial systems. Jaws closure to near-zero width indicates a complete miss, and closure at a value matching the object diameter suggests success. It requires no extra hardware and adds no meaningful cycle time. The problem, as established in Section 2.2.2, is that a nudge and a successful grasp produce the same width reading. Hosoda et al. [12] resolve this using wrist force-torque sensing to compare measured force profiles against expected ones, but that requires a dedicated six-axis sensor that is not part of this setup.

2.4.2 Direct Visual Verification via Binary Classification

Direct visual verification of grasp state by classifying the appearance of the gripper jaws has been explored in several forms. Morrison et al. [13] used a wrist-mounted depth camera to verify grasp success by comparing the expected and observed gripper-object geometry immediately after closure, though this approach is sensitive to occlusion by the gripper body and requires depth data at close range.

Amargant et al. [14] presents a two-stage architecture closely related to the method implemented in this project. YOLO-based detection localises the gripper in the frame, followed by a ResNet-based binary classifier determining whether the gripper is holding an object or empty. Their camera is mounted on the robot's head, further from the gripper than the wrist-mounted configuration used here. The 35-degree downward wrist angle adopted in this project is more informative as it captures the approach trajectory, the object

at the pre-grasp position, and the gripper state post-lift within the same frame, without requiring the head-mounted perspective distance or a separate gripper localisation step.

The YOLOv8n-cls binary classifier implemented here follows the same principle of direct gripper-state observation. After each grasp attempt, the robot lifts 200 mm and the camera captures a frame. The gripper region of interest is cropped and passed to the classifier, which outputs a confidence score for HOLDING and EMPTY. A confidence threshold of 0.8 is applied. Classifications below this threshold are treated as failed grasps to prevent false positives from propagating into the recovery logic. The classifier is trained on images collected at the exact wrist angle and lift height used in experiments, including partial jaw occlusion of the object.

2.4.3 Multi-Modal Sensor Fusion

Zhu et al. [2] show that combining visual and proprioceptive signals catches failures that neither modality catches alone: vision detects approach misalignment that pressure signals miss, and pressure detects in-hand slip that the camera cannot see through occlusion. The result is a substantially higher task completion rate than either signal individually.

In this project, sensor fusion is simpler. The DI8 binary force-resistant sensing, jaw width, and binary classifier output are combined. Gripper feedback is used as a first pass, followed by the binary classifier output confirming the final grasp state. This is justified by the hardware constraints. Gripper feedback alone generates false positives that the classifier resolves and the classifier alone cannot track attempt counts or timing, which the gripper signals provide.

2.5 Recovery Strategies

2.5.1 Re-Grasp After Position and Nudge Failures

The recovery for position and nudge failures is straightforward: re-localise the object and retry from the corrected position. When the classifier confirms a failed grasp, the object's current pixel position is transformed to robot-frame coordinates via the hand-eye calibration matrix and used as the new pick target. For a nudge specifically, this updated position is P2, the location the object was displaced to. Mahler et al. [15] describe the same principle in Dex-Net 2.0, where failed attempts inform subsequent ones, though their implementation uses a learned policy update rather than direct position correction.

2.5.2 Orientation Correction for Rigid Grippers

Eppner et al. [16] note that soft and underactuated grippers handle orientation variation naturally through compliance, and that rigid grippers must rely on explicit estimation and replanning instead. This project takes that a step further by treating orientation correction as a recovery action rather than something resolved at planning time. When an orientation failure is detected, the YOLOv8 OBB model outputs the in-plane rotation angle of the

cylinder directly, and the corrected gripper rotation is computed as $R_z = R_{z_default} + \text{OBB_angle} + 90$ degrees, aligning the jaws perpendicular to the cylinder long axis before re-attempting.

2.5.3 Slip Recovery

When slip is detected, the robot descends in a controlled manner to floor level, opens the gripper, re-detects the object, and re-grasps with increased RG2 grip force. Roberge et al. [18] show that tactile sensor arrays can detect the onset of slip before the object is fully lost, allowing grip force to increase proactively. This proactive capability is not achievable with simple FSR sensors on lightweight objects, so the recovery is purely reactive, the system waits for confirmed loss, then retry with a higher grip force.

2.6 Critical Synthesis and Research Gap

The most directly relevant prior work is Zhu et al. [2], which shows that combining visual tracking with proprioceptive gripper feedback across all phases of a pick-and-place cycle substantially improves autonomous failure handling compared to either signal alone. Their system is validated on real hardware and represents the current state of the art for closed-loop failure handling. The limitation is that it depends entirely on a pneumatic soft gripper with continuous pressure sensing, which has no equivalent in a rigid parallel-jaw gripper.

The closest work to the verification method implemented here is Amargant et al. [14], which uses a robot-mounted camera and a two-stage YOLO plus ResNet classifier to determine binary gripper state. This project builds on that by integrating the classifier output into a full failure classification and recovery state machine, and using a wrist-mounted downward-angle configuration eliminates the gripper localisation stage required by a head-mounted perspective.

Three gaps in the literature are addressed by this project. First, closed-loop failure detection and recovery for rigid parallel-jaw grippers, without wrist force-torque sensing or pneumatic proprioception. Second, orientation correction via OBB-guided gripper rotation. Third, a direct comparison between gripper-signal-only and camera-augmented verification across a structured 90-trial protocol provides a quantitative baseline that most existing work does not include.

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